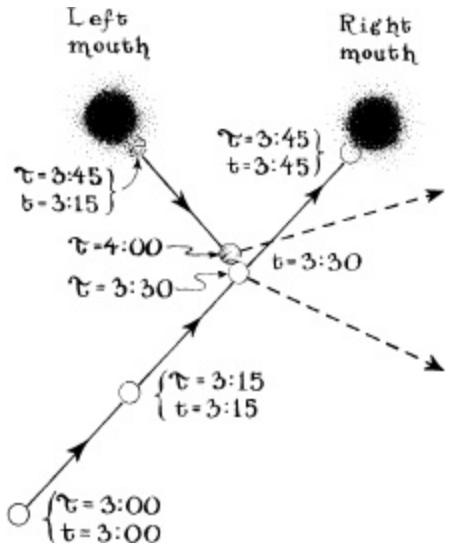
exceedingly accurate measurements on us; see Chapter 10), there is no way that either I or the billiard ball can go back in time and change our own histories.



14.9 Polchinski's billiard ball version of the matricide paradox. The wormhole is very short and has been made into a time machine, so that anything that enters the right mouth emerges, as measured on the outside, 30 minutes before it went in. The flow of time outside the mouth is denoted by the symbol *t;* the flow of time as experienced by the billiard ball itself is denoted by τ. The billiard ball is launched at *t* = 3:00 P.M. from the indicated location and with just the right velocity to enter the right mouth at *t* = 3:45. The ball emerges from the left mouth 30 minutes earlier, at *t* = 3:15, and then hits its younger self at *t* = 3:30 P.M., knocking itself off track so it cannot enter the right mouth and hit itself.

So what happens to the billiard ball? To find out, Morris, Yurtsever, and I focused our attention on the ball's *initial conditions*, that is, its initial location and velocity. We asked ourselves, "For the same initial conditions as led to Polchinski's paradox, is there any *other* billiard ball trajectory that, unlike the one in Figure 14.9, is a *logically self-consistent* solution to the physical laws that govern classical billiard balls?" After much discussion, we agreed that the answer was probably "yes," but we were not absolutely sure—and there was no time for us to figure it out. Morris and Yurtsever had completed their Ph.D.s and were leaving Caltech to take up postdoctoral appointments in Milwaukee and Trieste.

Fortunately, Caltech continually draws great students. There were two new ones waiting in the wings: Fernando Echeverria and Gunnar Klinkhammer. Echeverria and Klinkhammer took Polchinski's paradox and ran with it: